AN AI-BASED PREDICTIVE AND EXPLAINABLE WHOLE-LIFECYCLE MANAGEMENT SYSTEM FOR DISCUS FISH.

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Discus fish breeding is a high-sensitivity process that requires careful management during pre-spawning, egg, and fry stages. In practice, small-scale breeders, serious home breeders, and aquarium businesses often struggle to correctly interpret breeding-readiness signs, identify egg-stage risks, and manage fry development at the right time. Existing aquarium tools mainly support water monitoring, logging, or general fish care, but they do not provide integrated, Discus-specific breeding lifecycle guidance.

This proposal presents a hybrid vision-based and context-aware decision-support module for Discus breeding lifecycle management. The proposed methodology combines applied experimental research and design science. Pre-spawning readiness support will integrate breeder-reported behavioral and environmental observations with supportive analysis of adult Discus images. Egg-stage support will combine image classification of egg condition with timeline- and observation-based interpretation. Fry-stage support will rely mainly on lifecycle timing, breeder observations, warning logic, and recommendation rules. A rule-based fusion and explainable AI layer will combine model outputs, contextual inputs, and breeding-cycle history to generate stage-specific warnings and recommendations. The proposed implementation will use Python, PyTorch, OpenCV, FastAPI, Flutter, Firebase, and webcam-assisted image capture where available.

The expected outcome is a practical breeder-support tool that improves stage recognition, reduces avoidable egg and fry losses, and provides clearer breeder guidance than fragmented or generic aquarium tools. It is anticipated that the hybrid approach will be more suitable than image-only design because Discus breeding decisions depend on visual evidence, lifecycle timing, and breeder context together. Overall, the proposed module is expected to offer both academic value as a novel decision-support design and practical value as a niche breeder-support solution.

Keywords: Discus breeding, decision support system, computer vision, breeder support, ornamental aquaculture

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AGR	Abdominal Girth Ratio
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DSS	Decision Support System
FR	Functional Requirement
GPU	Graphics Processing Unit
LKR	Sri Lankan Rupee
ML	Machine Learning
NFR	Non-Functional Requirement
SDG	Sustainable Development Goal
UI	User Interface
WBS	Work Breakdown Structure
XAI	Explainable Artificial Intelligence

Table.0.0.1:List of Abbreviations Used in the Report

1. INTRODUCTION

1.1 Background and Context

Ornamental fish keeping and breeding are important both as a hobby and as a commercial activity. In Sri Lanka, the ornamental fish industry is an established export sector, and official industry reporting states that ornamental fish exports were valued at USD 24.46 million in 2024, while freshwater ornamental fish are mainly produced through small- and medium-scale captive breeding operations [1]. Recent government reporting also confirms that the sector contributes to foreign exchange earnings and employment through breeding, rearing, export, and retail activities [2]. Discus is also listed among the freshwater ornamental fish associated with Sri Lanka's export-oriented aquarium trade [1], [3]. At the broader market level, the ornamental fish industry is globally significant, with the Asia-Pacific ornamental fish market estimated at USD 1,433.8 million in 2023 and projected to grow further [4], [5].

Discus fish are a highly specialized ornamental freshwater cichlid that require more careful breeding and early-stage management than many common aquarium fish. Their breeding process is biologically sensitive because it involves substrate spawning, biparental egg guarding and fanning, temperature-dependent embryonic development, and early fry feeding on parental skin mucus [6]–[8]. As a result, Discus breeding is particularly vulnerable during the pre-spawning, egg, and fry stages, where incorrect decisions can lead to spawning failure, poor egg viability, fungal spread, parental stress, weak fry attachment, and early mortality [6]–[8].

This problem primarily affects small-scale Discus breeders, serious home breeders with dedicated breeding tanks, and specialized aquarium businesses. These users often need to determine the current breeding stage, identify warning signs, and decide what action should be taken next, yet existing support is usually fragmented or informal. Because Discus breeding depends on repeated observation across pre-spawning, egg, and fry stages, webcam-assisted image capture can support more consistent monitoring in breeder-managed tanks while still allowing practical user interaction through the application.In Sri Lanka, this issue has additional practical importance because industry sources already highlight the need for improved breeding technology, stronger technical knowledge, and better production support in ornamental aquaculture [2], [3].

Recent Discus-specific scientific evidence further shows that some commonly assumed breeder indicators are unreliable. Modern research indicates that external sexing of Discus from general appearance is unreliable, while early breeding success depends strongly on lifecycle timing, parental care, and environmental stability rather than on a single visible cue [7], [8]. Experimental evidence also shows that temperature significantly affects embryonic and larval development timing, meaning that egg- and fry-stage management cannot be treated as a fixed calendar process [8].

Although aquarium support tools exist, current solutions remain limited. In practice, breeders often rely on sensor-based monitoring systems, manual log applications, social media discussions, general-purpose fish-care apps, and personal trial-and-error. However, these solutions do not provide integrated support across pre-spawning, egg, and fry stages; they are usually generic rather than Discus-specific; and they do not combine image analysis, contextual user input, lifecycle timing, warning logic, and

stage-specific recommendations within a single workflow. As a result, breeders may recognize that something is wrong without clearly understanding what is happening or what action should be taken next.

To address this problem, this proposal introduces a hybrid vision-based and context-aware decision-support module for Discus breeding lifecycle management.

The proposed solution consists of three connected parts:

Model 1: pre-spawning readiness support**,** Model 2: egg-stage support, and a fry-stage support module. Together, these components combine image-based analysis with breeder observations and lifecycle timing in order to provide more practical and stage-aware support than a purely image-based system.

This proposal aligns primarily with SDG 8: Decent Work and Economic Growth**,** SDG 9: Industry, Innovation and Infrastructure, and SDG 14: Life Below Water by supporting breeding efficiency, applying digital technologies to aquaculture, and encouraging successful captive breeding that can reduce dependence on wild-caught specimens. It also aligns with the research areas of Artificial Intelligence and Machine Learning**,** Computer Vision and Image Analysis**,** Decision Support Systems**,** and Smart Aquaculture / Digital Agriculture Technologies.

Overall, Discus breeding remains a specialized and high-sensitivity problem affecting breeders, hobbyists, and ornamental fish businesses in Sri Lanka and in wider Discus-keeping communities. The lack of integrated, species-specific, and stage-aware support creates a clear need for the proposed decision-support module.

1.2 Literature Review

Recent scholarly work shows that artificial intelligence, computer vision, and smart aquaculture technologies are increasingly being applied to fish-related tasks such as detection, classification, behavior monitoring, welfare assessment, and automated management [9]–[13]. In particular, Li and Du [9] and An *et al.* [10] demonstrate that image-based and computer-vision approaches are already well established for fish classification, behavioral observation, and intelligent monitoring. More recent reviews further confirm that aquaculture is moving toward data-driven monitoring systems that combine computer vision, artificial intelligence, and contextual decision support [11]–[13].

However, the dominant focus of this literature remains commercial aquaculture rather than ornamental breeding support. Most existing studies address:

feeding optimization, biomass estimation, disease monitoring, welfare assessment, general fish tracking [9]–[13].

This is significant because it shows that the technical foundation already exists, but it has not yet been directed toward the breeding lifecycle management of sensitive ornamental species such as Discus.A second relevant research stream focuses on image-based fish condition and welfare assessment. Tamut *et al.* [14] show that convolutional neural networks can classify freshwater fish diseases from image data, while Fitzgerald *et al.* [15] demonstrate the increasing use of machine vision for non-invasive welfare monitoring. These studies support the use of image-based models for visually observable biological states, but they are not designed for pre-spawning readiness support, egg-condition interpretation, or fry-stage guidance.

The literature also makes clear that many fish-management decisions cannot be made using image evidence alone. Smart aquaculture reviews repeatedly emphasize the

value of combining visual data with contextual, environmental, and temporal information [11]–[13]. This is especially important in Discus breeding. Discus-specific studies show that external sexing is unreliable**,** parental mucus feeding is biologically central, and embryonic and larval development are strongly affected by temperature [7], [8], [16]. These findings imply that some breeding stages can be partially observed visually, while others require **t**imeline information, breeder observations, and environmental context.

In addition to scholarly literature, currently available aquarium-support products remain limited in relation to the proposed problem. Tools such as Seneye and Neptune Apex / Apex Fusion provide useful water monitoring, alerts, and tank-control functions, but they do not provide Discus-specific breeding-stage interpretation [17], [18]. Therefore, both the research literature and the current solution landscape indicate a clear need for a specialized, stage-aware, and explainable decision-support module for Discus breeding.

Table 1.1: Summary of Reviewed Literature and Existing Solutions

Category	Main Focus	Key Limitation	Relevance to Proposed Research
AI and computer vision in aquaculture [9]–[13]	Fish monitoring, feeding, welfare, detection, and smart aquaculture management	Mainly focused on commercial aquaculture rather than sensitive ornamental breeding	Supports the technical feasibility of AI and computer vision methods
Image-based fish condition analysis [14], [15]	Disease detection and visually observable biological conditions	Not designed for Discus pre-spawning support, egg-condition interpretation, or fry-stage care	Supports the use of image analysis for visible biological states
Discus-specific reproductive biology [7], [8], [16]	Parental care, developmental timing, mucus feeding, and reproductive constraints	Strong biological insight, but no integrated AI-based breeding support design	Supports the biological basis for the proposed hybrid module
Seneye [17]	Water-quality monitoring, temperature alerts, and aquarium condition tracking	Sensor-focused only; does not provide Discus breeding-readiness assessment, egg-condition analysis, or fry-stage decision support	Shows the limitation of environment-only monitoring
Neptune Apex / Apex Fusion [18]	Smart aquarium automation, remote monitoring, and equipment control	Focuses on tank automation and parameter management rather than species-specific breeding lifecycle guidance	Shows that tank control is not the same as breeding decision support

1.3 Research Gap

The reviewed literature shows that artificial intelligence, computer vision, and smart aquaculture systems are increasingly used for fish-related tasks such as detection, classification, behavior monitoring, welfare assessment, and automated management [9]–[15]. However, this work is still focused mainly on commercial aquaculture rather than the breeding lifecycle management of sensitive ornamental fish species. As a result, the practical breeding-support needs of Discus keepers and breeders remain insufficiently addressed.

The first major gap is a species-specific and lifecycle-specific gap. No peer-reviewed study identified in this review presents an integrated Discus-specific decision-support approach covering pre-spawning readiness support, egg-stage support, and fry-stage care guidance within a single module. Existing studies mainly address isolated tasks such as fish detection, disease classification, feeding analysis, welfare monitoring, and general fish tracking [9]–[15]. These are technically useful, but they do not solve the breeder's core problem of managing a connected sequence of sensitive stages from breeding preparation to eggs and then to fry.

The second major gap is a methodological gap. Current image-based fish-analysis approaches show that computer vision is feasible in aquatic environments, but they also face practical limitations such as variable lighting, reflections, occlusion, unstable image quality, and limited generalization [9], [12], [13]. In Discus breeding, some cues can be visually observed, but many important decisions cannot be made from image evidence alone. Discus-specific evidence shows that external sexing is unreliable**,** early development is strongly influenced by temperature and timing, and fry feeding on parental mucus is biologically central [7], [8], [16]. This indicates that a purely image-based approach is insufficient, while a purely manual or log-based tool is also insufficient because it does not utilize available visual evidence. Therefore, a hybrid vision-based and context-aware approach is required.

The third gap exists in currently available products and services. Tools such as Seneye and Neptune Apex / Apex Fusion provide useful support for water monitoring, alerts, and aquarium automation, but they do not provide Discus-specific breeding lifecycle interpretation [17], [18]. These systems help manage the aquarium environment, but they do not answer breeder questions such as whether a pair is showing meaningful pre-spawning readiness, whether eggs are visually healthy or at risk, or what actions should be taken during fry-stage progression.

The novelty of this proposal lies in how it addresses these gaps through a function-specific hybrid design. The proposed individual function does not force artificial intelligence onto every breeding stage. Instead, it applies different methods where they are most appropriate:

Model 1 uses breeder-reported behavioral and environmental observations together with supportive image-based analysis for pre-spawning readiness support.

Model 2 combines egg-condition image classification with timeline- and context-based egg-stage interpretation.

The fry-stage support module uses lifecycle timing, breeder observations, warning logic, and recommendation rules rather than unreliable fry image classification.

These components are connected through warning generation, evidence-consistency checks, and stage-specific recommendations, creating a more practical and biologically grounded system than existing isolated tools.
Accordingly, the core research gap addressed by this proposal can be stated as follows: current research and existing aquarium-support systems do not provide an integrated, species-specific, hybrid vision-based and context-aware decision-support module for Discus breeding lifecycle management that combines pre-spawning support, egg-condition analysis, egg-stage guidance, and fry-stage care support in an explainable and practically deployable form.
Table 2 summarizes the identified research gap by comparing the limitations of current aquarium-support applications and selected scholarly approaches with the contribution of the proposed module.

Table 1.2 :Research Gap Based on Existing Mobile/Support Solutions and Related Research

Aspect	Current App / Research Example	Limitation	Proposed Contribution
Aquarium monitoring	Seneye [17]	Focuses on water quality, temperature, and alerts only; does not interpret Discus breeding stages	Adds biological stage interpretation and breeder guidance beyond environment monitoring
Smart tank automation	Neptune Apex / Apex Fusion [18]	Supports tank automation and parameter control, but not Discus-specific breeding lifecycle support	Introduces species-specific breeding, egg, and fry-stage decision support
Manual aquarium tracking	Discus AquaBook [19]	Supports logging of pairs and offspring, but lacks image analysis, warning logic, and intelligent recommendations	Extends Discus-focused record keeping into a hybrid decision-support system
AI in aquaculture research	Li and Du [9]	Demonstrates computer vision feasibility, but mainly for commercial aquaculture tasks rather than ornamental breeding	Applies AI selectively to Discus breeding where visual evidence is actually useful
Fish condition analysis research	Tamut *et al.* [14]	Supports image-based biological condition classification, but not Discus breeding lifecycle management	Uses image classification for egg condition while avoiding weak image-only claims in other stages
Smart aquaculture decision support	Huang and Khabusi [13]	Supports combining AI with contextual and environmental data, but not in a Discus-specific breeding workflow	Builds a hybrid vision-based and context-aware module tailored to Discus pre-spawning, egg, and fry stages

2. OBJECTIVE

2.1 Main Objective

The main objective of this individual research is to develop and evaluate a hybrid vision-based and context-aware decision-support module for Discus pre-spawning, egg, and fry-stage management using breeder-provided visual input, contextual observations, lifecycle timing, warning logic, and stage-specific recommendations in order to support more accurate and timely breeder decision making.

2.2 Specific Objectives

1. To achieve the main objective, the following specific objectives are defined for this individual function:

2. To develop a pre-spawning readiness support component that combines breeder-provided behavioral and environmental observations with supportive analysis of adult Discus images to assess breeding readiness and recommend appropriate next actions.
3. To design an egg-condition assessment component that classifies Discus egg clusters into visually meaningful categories such as healthy, unhealthy, mixed clutch, or fungal infection visible in order to support early identification of egg-stage risks.
4. To develop an egg-stage interpretation component that uses lifecycle timeline information and breeder observations to determine likely egg-stage progression and generate stage-appropriate warnings and guidance.
5. To implement a fry-stage care support component that uses lifecycle timing, breeder observations, and warning-rule analysis to guide Discus fry management through attachment, free-swimming, feeding transition, and early separation-related stages.
6. To integrate image-based outputs, contextual inputs, timeline logic, and warning rules into a unified decision-support layer capable of generating actionable recommendations, caution alerts, and stage-specific care guidance for breeders.
7. To evaluate the proposed image-based and logic-based components using appropriate validation methods in order to assess their performance, interpretability, and practical usefulness for Discus breeding lifecycle support.

Table 2.1:Alignment of Objectives with Proposed Individual Components

Specific Objective	Related Component	Expected Output
Develop pre-spawning readiness support	Model 1	Readiness support level with contextual explanation and next-step guidance
Design egg-condition assessment	Model 2A	Egg condition classification result
Develop egg-stage interpretation	Model 2B / Egg Stage Logic	Egg-stage guidance based on timeline and context
Implement fry-stage care support	Fry Support Module	Fry-stage guidance and lifecycle-based recommendations
Integrate outputs into decision-support layer	Fusion / Warning / Recommendation Layer	Actionable recommendations, caution alerts, and warning flags
Evaluate models and overall module	Evaluation Component	Performance, feasibility, and interpretability assessment

3. METHODOLOGY

3.1 Research Approach

This individual research adopts an applied experimental research approach combined with a design science-oriented system development perspective. The experimental aspect is appropriate because the proposed module contains image-based components that must be trained, compared, and evaluated using measurable performance metrics. The design science aspect is also necessary because the final output is not only a predictive model, but a hybrid decision-support module that integrates image analysis, breeder-provided contextual inputs, lifecycle timing, warning logic, and recommendation generation into a practical breeder-support system.
This approach is suitable because the Discus breeding problem cannot be solved by image analysis alone. Some stages contain useful visual evidence, while others depend more strongly on breeder observations, environmental conditions, and timeline information. Therefore, the methodology combines model-based analysis with structured decision-support logic.

3.2 Technical Methods and Algorithms

The proposed methodology is organized into three main functional components:

- Model 1 – Pre-spawning readiness support
- Model 2 – Egg-stage support
- Fry-stage support module

The overall workflow begins with breeder-selected stage input through the system interface. The breeder first selects whether support is required for pre-spawning, egg-stage, or fry-stage care. The system then requests the relevant external webcam-acquired image or frame together with stage-specific contextual questions. Captured images are passed through an image quality validation stage before analysis. The use of a fixed external webcam improves image consistency by reducing uncontrolled variation in distance, angle, and framing, making it more suitable for repeated observation in dedicated breeding tanks than ad hoc handheld image capture.

Model 1 – Pre-spawning readiness support
This component is designed as a context-prioritized hybrid support model. It combines breeder-reported observations such as pair bonding, surface-cleaning behavior, territorial aggression, environmental readiness, and breeding history with supportive image-based analysis of adult Discus body shape. The image component uses EfficientNet-B0 as the primary model and ResNet18 as the baseline comparison model. The output is a breeding-readiness support level with next-step guidance rather than a hard biological claim.



Figure 3.1:Adult Discus Images Used for Supportive Pre-Spawning Visual Analysis

Model 2 – Egg-stage support



Figure3.2:Example Egg-Condition Classes Used in the Proposed Egg-Stage Support Component

This component contains two parts:
Egg Condition Classifier:
Analyzes the egg-cluster image and classifies it as:

- healthy / likely viable,
- unhealthy / likely non-viable,
- mixed clutch,
- fungal infection visible.

EfficientNet-B0 is the primary model, and ResNet18 is used as the baseline comparison model.

Egg Stage Interpretation Logic:
Uses non-image information such as elapsed time since spawning, parental fanning behavior, white-egg removal, temperature, disturbance history, and wriggler visibility. This design is informed by evidence showing that temperature significantly affects embryonic and larval development in *Symphysodon aequifasciatus* [16].
Fry-stage support moduleThe fry-stage component is implemented as a timeline- and observation-driven decision-support module rather than a primary deep-learning classifier. This is because key fry-stage decisions depend more strongly on days since hatching, free-swimming timing, attachment to parents, feeding response, parent behavior, and water-management practices. An optional binary visual confirmation check is included to record whether fry are visibly clustered on or near parental flanks.

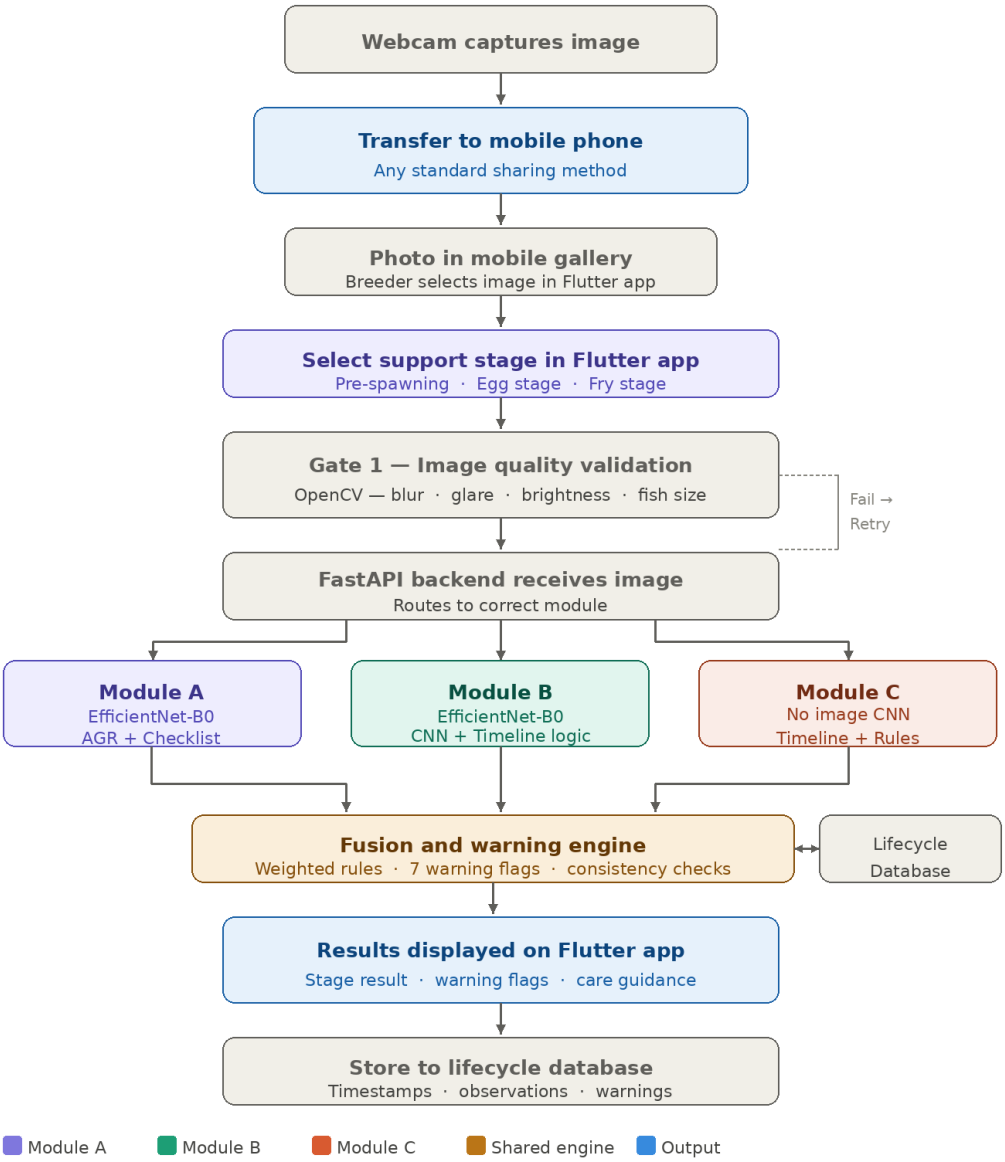


Figure 3.3: User Workflow of the Proposed Discus Breeding Lifecycle Support Module

3.2.1 Fusion and explainable AI implementation

The fusion and warning engine will be implemented as a rule-based decision-support layer that combines outputs from image-based models, breeder-provided observations, lifecycle timing variables, and stored breeding-cycle history into one final stage-specific interpretation. First, model outputs and breeder inputs will be converted into structured variables such as readiness support level, egg-condition class, elapsed time since spawning, attachment status, or feeding response. These variables will then be evaluated through if-then rules and threshold-based conditions to generate warning flags, interpret the current stage, and recommend appropriate next actions.

Explainable AI will be supported in two ways. First, the system will provide decision-level reasoning by showing which contextual conditions or lifecycle rules triggered

the final warning or recommendation. Second, for Model 1, simple morphology-based support features from the adult Discus image will be used alongside the classifier output to make the pre-spawning assessment more understandable. This allows the system to provide explainable breeder guidance rather than raw model predictions only.

3.3 Tools and Platforms with Justification

- Python is used as the main programming language because it provides strong support for machine learning, computer vision, and backend development.
- PyTorch is used as the deep learning framework because it supports transfer learning and model experimentation.
- OpenCV is used for image preprocessing, image validation, and extraction of simple visual features from captured images.
- FastAPI is used for backend integration because it is lightweight and suitable for serving inference and decision-support outputs.
- Flutter is selected as the user interface framework because it supports flexible cross-platform development and practical breeder interaction.
- Firebase is used as the cloud-backed data storage platform for breeding-cycle records, including spawning timestamps, hatch dates, observations, warning events, and recommendation history. This storage is architecturally necessary because the egg-stage interpretation and fry-stage guidance components depend on previously recorded lifecycle events across multiple sessions.
- Webcam-assisted image capture is supported as a practical way to improve observation consistency in breeder-managed setups.

3.4 Data Collection and Ethical Considerations

Different components require different forms of data.

- Model 1 requires side-profile adult Discus images together with breeder-provided behavioral and environmental observations.
- Model 2A requires egg-cluster images representing healthy, unhealthy, mixed, and fungal-infection cases.
- Model 2B requires non-image data such as spawning time, parental egg-care behavior, disturbance history, and temperature records.
- The Fry Module requires mainly non-image data including hatch timing, free-swimming timing, attachment behavior, feeding response, parent behavior, survival observations, and water-management practices.

Data can be collected from breeder-owned tanks, controlled aquarium observations, structured breeder records, and carefully selected breeder-shared materials. Webcam-assisted capture may improve consistency in repeated observation, but the broader system remains centered on practical breeder-supported input collection.

This research does not require invasive experimentation on fish. Data collection should occur during routine care and normal breeding observation. Ethical care must be taken to ensure that:

- fish are not intentionally stressed for image capture,
- breeder-provided images and records are handled responsibly,
- stored data is protected appropriately,
- system outputs are presented as guidance rather than guaranteed biological outcomes.

3.5 Validation Strategy and Performance Metrics

The image-based components will be validated using standard supervised learning evaluation methods.
For Model 1 supportive image analysis and Model 2A egg condition classification, the dataset will be divided into:

- training set, validation set, testing set,or alternatively evaluated using stratified cross-validation if dataset size is limited.
- The main performance metrics will include: accuracy, precision, recall, F1-score, confusion matrix.Where class imbalance exists, macro F1-score will be emphasized in addition to raw accuracy.
- The evaluation plan will include:
- Baseline comparison
  ResNet18 vs EfficientNet-B0
- Ablation-style comparison
  image-only output vs hybrid output with contextual integration
- Practical validation
  whether the final outputs are interpretable, useful, and actionable for breeder decision making

For Model 2B egg-stage interpretation and the Fry Module, validation will focus on: logical consistency,stage-interpretation correctness,warning correctness,scenario-based rule validation,practical suitability of generated recommendations.

3.6 Methodology Summary

Overall, the proposed methodology uses a hybrid structure in which image-based analysis is applied where visual evidence is meaningful, while contextual reasoning, lifecycle timing, and rule-based logic are used where image-only inference would be weak. This makes the proposed individual module technically justified, scientifically defensible, and feasible within the scope of a final-year research project.

Table 3.1:Summary of Technical Methods Used in Each Component

Component	Main Method	Supporting Method	Output
Model 1	Context-prioritized decision support	Supportive image analysis + breeder inputs	Breeding-readiness support level
Model 2A	EfficientNet-B0 classification	Contextual fusion	Egg condition result
Model 2B / Egg Stage Logic	Rule-based lifecycle interpretation	Timeline + breeder observations	Egg-stage guidance and interpretation
Fry Module	Rule-based lifecycle logic	Warning analysis + observation inputs	Fry-stage care guidance

Table 3.2:Validation Metrics and Evaluation Strategy

Component	Validation Method	Metrics
Model 1 image support	Train/validation/test split or cross-validation	Accuracy, Precision, Recall, F1-score, Confusion Matrix
Model 2A	Train/validation/test split or cross-validation	Accuracy, Precision, Recall, F1-score, Confusion Matrix
Model 2B / Egg Stage Logic	Rule and scenario-based validation	Stage consistency, rule correctness, interpretation usefulness
Fry Module	Logic and output consistency evaluation	Rule correctness, stage consistency, recommendation usefulness

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed individual function is designed as a hybrid vision-based and context-aware decision-support module within the broader AI-powered ornamental fish breeding and care system. This individual proposal focuses specifically on the Discus breeding lifecycle support subsystem, which covers pre-spawning readiness support, egg-stage support, and fry-stage care guidance. The architecture combines image-based analysis, timeline-based reasoning, breeder-provided contextual inputs, fusion logic, and breeding-cycle record storage within one modular workflow.

4.2 System Diagram of Components

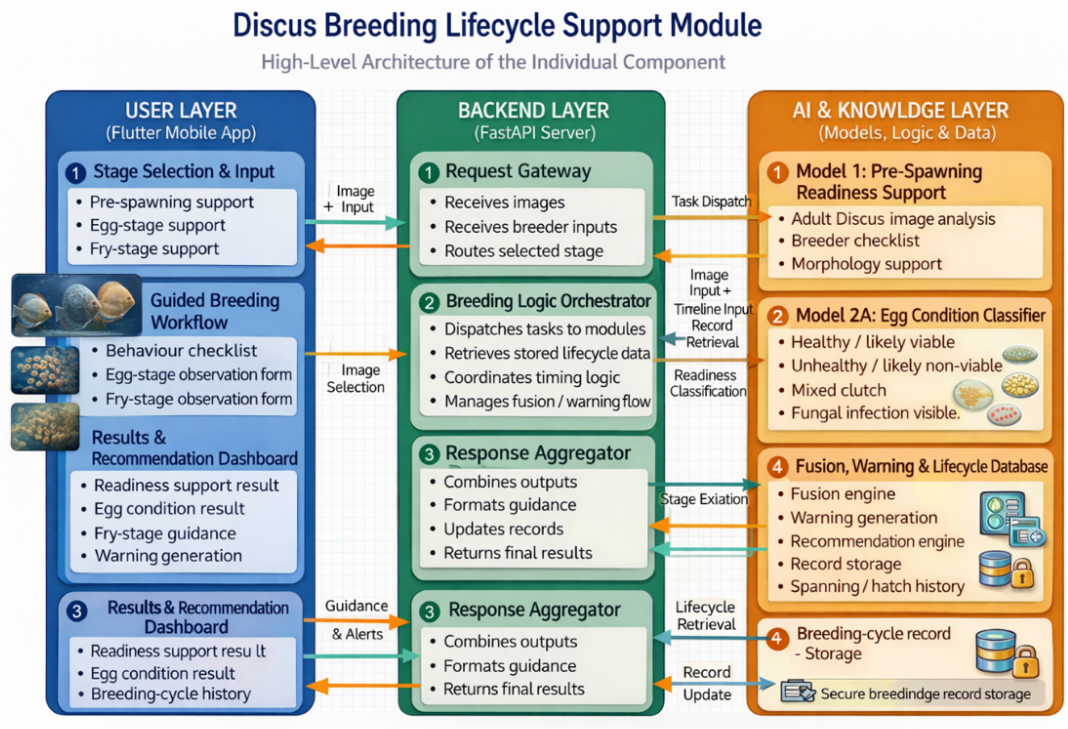


Figure 4.1:High-Level System Architecture of the Proposed Individual Module

4.3 Highlighting the Individual Component and Its Parts

Within the broader group system, the individual component proposed in this research is the Discus breeding lifecycle support subsystem. This subsystem contains the following main parts:

- Model 1: Pre-spawning readiness support
- Explainable morphology support layer
- Model 2A: Egg condition classifier
- Model 2B: Egg stage interpretation logic
- Fry-stage support module

- Fusion, warning, and recommendation engine
- Breeding-cycle data storage

This structure shows that the individual function is not a single classifier, but a connected subsystem that combines image-based and non-image-based reasoning.

4.4 Component Interactions and Data Flow

The architecture combines image-based analysis, timeline-based reasoning, breeder-provided contextual inputs, fusion logic, and breeding-cycle record storage within one modular workflow.

1. The breeder selects the required support stage through the application interface.
2. The application sends images, observations, and lifecycle inputs to the backend API.
3. The backend performs image validation and retrieves any relevant breeding-cycle history from Firebase.
4. The validated input is routed to the relevant internal component:
5. Model 1 for pre-spawning readiness support
6. Model 2A and Model 2B for egg-stage support
7. Fry-stage support module for fry care guidance
8. Intermediate outputs are integrated through the fusion, warning, and recommendation engine.
9. Final outputs are generated in the form of:
10. interpreted stage status,
11. warning flags,
12. recommended next actions.
13. The results are returned to the application and displayed to the user.
14. New observations, timestamps, warnings, and outputs are stored in Firebase breeding-cycle records for continuity across sessions.

4.5 Scalability and Security Considerations

The proposed architecture is modular and scalable, allowing the pre-spawning, egg-stage, and fry-stage components to be improved independently without redesigning the whole system. The use of a backend-supported structure also allows future updates to models, rules, and recommendation logic while keeping the breeder-facing interface stable.

Basic security considerations are also necessary. Since the system may store breeder-uploaded images, lifecycle records, warning history, and recommendation data, the architecture should ensure:

protected communication between the frontend interface, webcam-assisted capture workflow, and backend,

secure storage of breeder records,

controlled access to breeding-cycle history,

appropriate handling of uploaded images and user-linked outputs.
Because the module functions as a breeder-support tool rather than an autonomous biological controller, outputs should be clearly presented as advisory rather than guaranteed outcomes.

Table 4.1:Summary of High-Level Architecture Components

Component	Main Role	Key Input	Key Output
	Collect breeder inputs and display results	Images, observations, lifecycle data	User-facing outputs
Backend API	Route and coordinate processing	Mobile requests	Routed tasks and returned responses
Image Validation Component	Check image suitability	Adult fish image / egg image	Valid image or rejection message
Model 1	Pre-spawning readiness support	Adult image + breeder context + pair history	Readiness support level
Explainable Morphology Layer	Extract interpretable body-shape evidence	Adult Discus image	AGR and contour-based support features
Model 2A	Egg condition classification	Egg-cluster image	Egg condition result
Model 2B	Egg-stage interpretation	Timeline + breeder observations + spawning history	Egg-stage guidance
Fry-Stage Support Module	Fry-stage support	Lifecycle data + fry observations	Fry-stage guidance
Fusion, Warning, and Recommendation Engine	Integrate evidence and generate decision support	Outputs from models + context + stored history	Warnings and recommendations
Data Storage	Store and retrieve breeding-cycle history	Timestamps, observations, results	Stored records and retrieved context

5. USER REQUIREMENTS

The proposed individual module is intended to support breeders and Discus keepers during the most sensitive stages of the breeding lifecycle. Since the system functions as a decision-support module rather than a fully autonomous control system, its requirements focus on how users provide images, observations, and lifecycle data, and how the system returns interpretable results, warnings, and recommendations.

5.1 Stakeholder Identification

The main stakeholders of the proposed module are:

- Small-scale Discus breeders who need support for pre-spawning assessment, egg-stage interpretation, and fry survival management.
- Serious home breeders who maintain stable Discus breeding tanks and can benefit from repeated webcam-assisted monitoring.
- Aquarium shops and specialized ornamental fish businesses that keep or support Discus stock and may benefit from stage-specific breeder guidance.
- Future developers and maintainers who may extend the module, improve models, or adapt the system for broader use.

Table 5.1:Stakeholder Summary

Stakeholder	Role	Main Interest
Small-scale Discus breeders	Primary user	Reduce breeding, egg, and fry-stage losses
Serious home breeders	Primary user	Obtain practical stage-specific guidance
Aquarium shops / ornamental fish businesses	Secondary user	Improve breeding support and stock management
Future developers / maintainers	Secondary stakeholder	Extend and improve the module over time

5.2 Functional Requirements

Table 5.2:Functional Requirements of the Proposed Module

ID	Functional Requirement
FR1	The system shall allow the user to select the required support stage: pre-spawning, egg-stage, or fry-stage.
FR2	The system shall allow the user to upload a side-profile adult Discus image for pre-spawning support and an egg-cluster image for egg-condition assessment.
FR2A	The system shall support repeated stage-relevant image/frame capture from a fixed external webcam positioned near the breeding tank.
FR3	The system shall validate uploaded images for blur, brightness, glare, and subject visibility before analysis.
FR4	The system shall allow the user to enter behavioral, environmental, and lifecycle observations relevant to the selected stage.

FR5	The system shall generate a pre-spawning readiness support result by combining breeder-reported observations with supportive image-based analysis.
FR6	The system shall classify egg images into healthy / likely viable, unhealthy / likely non-viable, mixed clutch, or fungal infection visible.
FR7	The system shall interpret egg-stage progression using timeline data and breeder observations rather than image classification alone.
FR8	The system shall provide fry-stage guidance using lifecycle timing, breeder observations, and warning logic.
FR9	The system shall generate warning flags and stage-specific recommendations instead of returning raw model outputs only.
FR10	The system shall store and retrieve breeding-cycle records such as spawning time, hatch timing, observations, warnings, and previous recommendations to support continuity across sessions.

5.3 Non-Functional Requirements

Table 5.3:Non-Functional Requirements of the Proposed Module

ID	Non-Functional Requirement
NFR1	The module shall provide a simple and breeder-friendly mobile interaction flow for image upload, stage-specific input, and result viewing.
NFR2	The module shall request only the inputs relevant to the currently selected stage in order to reduce user confusion.
NFR3	The module shall return results within a reasonable response time suitable for practical breeder use.
NFR4	The module shall reject low-quality images rather than generating unreliable outputs from unsuitable visual input.
NFR5	The module shall provide consistent stage interpretation and clearly indicate warning conditions or evidence conflicts where uncertainty exists.
NFR6	The module shall protect breeder-uploaded images, lifecycle records, and stored breeding-cycle history from unauthorized access.
NFR7	The module shall be modular so that the pre-spawning, egg-stage, and fry-stage components can be improved independently.
NFR8	The module shall provide explainable outputs in the form of interpreted status, warning flags, and recommended next actions rather than raw predictions alone.
NFR9	The module shall remain feasible for implementation within the scope of an undergraduate final-year project while still supporting future extension.

6. COMMERCIALIZATION PLAN

The proposed module has commercialization potential because it addresses a recurring and costly problem in Discus breeding: breeders often struggle to correctly interpret pre-spawning readiness, egg condition, and fry-stage care requirements during the most sensitive lifecycle stages. The proposed solution is better positioned as a niche breeder-support product for Discus breeding setups rather than as a general aquarium application.

6.1 Target Market, Value Proposition, Revenue, and Pricing

Table 6.1:Commercialization Summary of the Proposed Module

Aspect	Description
Target Market	Small-scale Discus breeders, serious home breeders with dedicated breeding tanks, aquarium shops, and specialized ornamental fish businesses
Customer Persona	A breeder who manages one or more Discus breeding tanks, wants to reduce egg and fry losses, and needs repeated stage-specific guidance rather than general aquarium advice
Value Proposition	A Discus-specific, stage-aware, and explainable breeder-support module that combines webcam-based image analysis, breeder observations, lifecycle timing, warnings, and actionable recommendations
Revenue Model	Freemium model with a free basic version and premium features for advanced analysis, breeding-cycle history, and detailed warning support
Pricing Strategy	Low-cost monthly and annual plans for individual breeders, with a higher-tier package for shops or multi-tank users
Competitive Advantage	Discus-specific focus, hybrid vision- and context-aware design, breeding lifecycle coverage across pre-spawning, egg, and fry stages, and explainable outputs rather than generic aquarium advice
Intellectual Property Considerations	Protectable elements include application source code, trained models, curated datasets, fusion and warning logic, recommendation workflow, and future brand identity

Table 6.2:Proposed Pricing Structure for the Commercial Version

Plan	Price
Free Tier	LKR 0
Premium Monthly	LKR 499/month
Premium Annual	LKR 4,500/year

6.2 Cost Estimation

The following estimate represents a small early-stage commercialization cost for deploying the module as a niche breeder-support product. This is separate from the academic project development budget in Section 7.

Table 6.3:Estimated Early Commercialization Cost

Cost Item	Estimated Cost (LKR)
App refinement and UI improvement	20,000
Backend hosting and API deployment	20,000
Model maintenance and retraining	25,000
Firebase storage and backup	10,000
Testing and quality assurance	15,000
User support and documentation	10,000
Marketing / breeder community awareness	15,000
Optional webcam setup support	12,000
Contingency	15,000
Total Estimated Cost	**142,000**

6.3 COMMERCIALIZATION SUMMARY

Overall, the proposed module is commercially promising as a niche breeder-support product rather than a mass-market aquarium application. Its strongest advantage is providing Discus-specific breeding lifecycle support through image-assisted analysis, contextual reasoning, breeding-cycle continuity, and stage-based recommendations. A freemium-to-premium model is the most realistic commercialization path, with future expansion possible through advanced breeder features and adaptation to other sensitive ornamental fish species.

7. BUDGET AND JUSTIFICATION

The budget for this individual module covers the essential resources required for designing, developing, testing, and documenting the proposed Discus breeding lifecycle support system. Since the module is developed in an academic setting, the budget remains practical by relying on open-source tools wherever possible. The main costs are associated with hardware support, cloud services, data handling, and limited operational expenses required for development and validation.

Table7.1:Estimated Budget and Justification for the Proposed Individual Module

Budget Item	Category	Estimated Cost (LKR)	Justification
Personal laptop/computer usage	Hardware	0	Existing personal device will be used for development, testing, and reporting
External storage / backup	Hardware	3,000	Required to store datasets, trained models, and project files safely
USB webcam	Hardware	6,000	Supports repeated stage-relevant image capture during development and testing
Open-source software tools	Software Licenses	0	Python, PyTorch, OpenCV, FastAPI, Flutter, and related tools do not require license fees
Cloud compute / GPU support	Cloud Services	8,000	Needed for model training and testing if local hardware is insufficient
Backend hosting / API deployment	Cloud Services	6,000	Required for testing backend-based inference and application integration
Firebase / cloud data storage	Cloud Services	4,000	Needed to store breeding-cycle records, warnings, recommendation history, and lifecycle data
Internet and data usage	Operational	5,000	Needed for literature access, cloud usage, collaboration, and data transfer
Data collection / breeder consultation	Operational	5,000	Supports limited breeder consultation and practical data gathering
Printing and documentation	Documentation	3,000	Required for report preparation and submission materials
Contingency	Miscellaneous	5,000	Covers minor unexpected technical or operational expenses
Total Estimated Budget		**45,000**	

8. WORK BREAKDOWN STRUCTURE (WBS)

The overall project workload is divided into four functional streams so that each team member is responsible for a distinct technical area of the broader ornamental fish breeding and care system. This supports balanced workload distribution, parallel development, and clearer ownership of outputs. Within this structure, the present individual report focuses on Member 3, whose responsibility is the Discus breeding lifecycle support module covering pre-spawning readiness support, egg-stage support, fry-stage care guidance, and fusion-based recommendation generation.

The task flow of the individual component follows a logical sequence from literature and scope definition to model design, rule development, integration, and evaluation. This progression ensures that the final module is developed in a structured and traceable manner.

Table 8.1:Work Breakdown Structure and Team-Level Task Distribution

WBS ID	Member / Component	Key Tasks	Timeline Alignment	Workload Distribution
1.0	Member 1 – Water Quality Monitoring Module	Define water-quality parameters, monitoring logic, alerts, and frontend/backend integration	Early to mid project	Independent environmental-monitoring stream
2.0	Member 2 – Fish Disease Analytics Module	Collect disease data, design classification workflow, implement disease-analysis logic, and integrate outputs	Early to mid project	Independent AI-based fish-health stream
3.0	Member 3 – Discus Breeding Lifecycle Support Module	Design and implement pre-spawning support, egg-stage support, fry-stage guidance, fusion logic, and recommendation output	Throughout project, with strongest focus in the middle phase	Hybrid AI and decision-support stream
3.1	Member 3 – Literature review and scope definition	Review Discus breeding biology, identify the research gap, define objectives, and finalize module scope	Early phase	Academic and conceptual foundation
3.2	Member 3 – Input and workflow design	Design user workflow, stage-specific inputs, image input process, and	Early phase	Functional design of breeder interaction

		contextual questionnaires		
3.3	Member 3 – Model 1 development	Develop pre-spawning readiness support using breeder context and supportive image analysis	Early to mid phase	Core readiness-support component
3.4	Member 3 – Model 2 development	Develop egg-condition classification and egg-stage interpretation logic	Mid phase	Core egg-stage support component
3.5	Member 3 – Fry module development	Develop fry-stage lifecycle rules, observation inputs, care guidance, and warning triggers	Mid to late phase	Rule-based fry-stage support component
3.6	Member 3 – Fusion and explainability layer	Integrate outputs from all subcomponents, apply warning logic, and generate stage-specific recommendations with explainable reasoning	Late phase	Final decision-support layer
3.7	Member 3 – Validation and refinement	Define validation criteria, test image and rule-based components, and refine outputs	Late phase	Final technical evaluation and improvement
4.0	Member 4 – Feeding Management and Tank Density Alert Module	Implement feeding estimation, biomass-related support, density alerts, and integration support	Early to mid project	Parallel contribution to the overall system
5.0	Group Integration	Combine all member modules, test complete system flow, and prepare final demonstration and presentation	Late project phase	Shared team responsibility

9. GANTT CHART

The following Gantt chart presents the semester-wise schedule for the proposed individual function, aligned with the official Research Project timeline for the 2026 Regular Batch. The activities are organized by week to show how the Discus breeding lifecycle support module will progress from proposal preparation through implementation, evaluation, and final delivery.

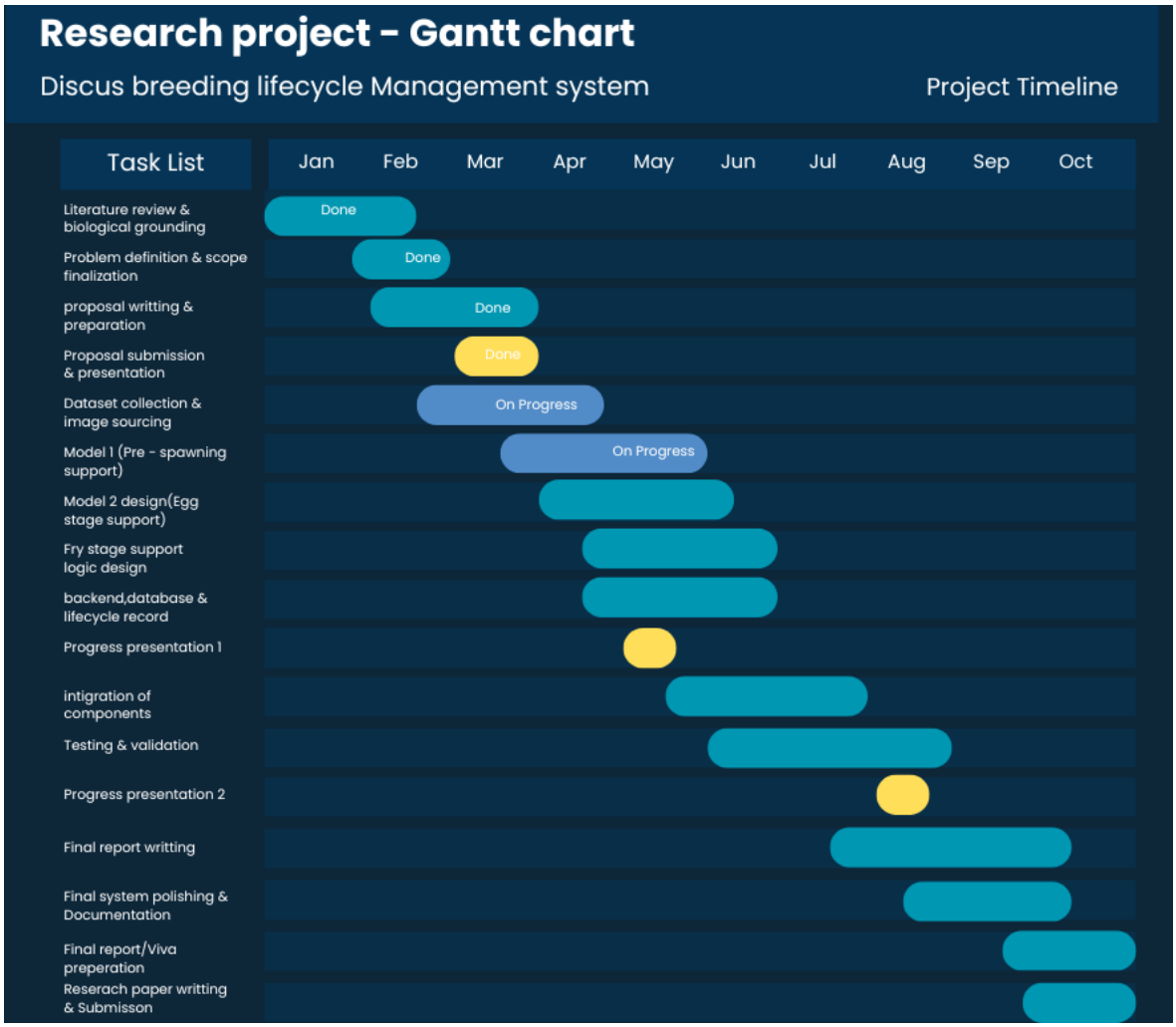


Figure 9.1:Gantt chart

REFERENCE LIST

[1] Sri Lanka Export Development Board, *Industry Capability Report: The Ornamental Fish Industry*. Colombo, Sri Lanka, 2025. Available: https://www.srilankabusiness.com/ebooks/industry-capability-report-ornamental-fish-industry-2025.pdf (Accessed: Mar. 19, 2026).

[2] Ministry of Fisheries, Aquatic and Ocean Resources, Sri Lanka, "Progress Report 2025," Colombo, Sri Lanka, 2025. [Online]. Available: https://www.fisheries.gov.lk/web/images/downloads/pdfs/2025/Progress_Report_-_2025_English_compressed.pdf (Accessed: Mar. 19, 2026).

[3] Sri Lanka Export Development Board, "Export-Oriented Ornamental Fish Farming in Sri Lanka," Colombo, Sri Lanka, Sep. 2020. [Online]. Available: https://www.srilankabusiness.com/blog/export-ornamental-fish-sri-lanka.html (Accessed: Mar. 19, 2026).

[4] Grand View Research, *Asia Pacific Ornamental Fish Market Size, Share and Trends Analysis Report*, 2024. Available: https://www.grandviewresearch.com/industry-analysis/asia-pacific-ornamental-fish-market-report

[5] J. H. Peh and M. N. Azra, "A global review of ornamental fish and shellfish research," *Aquaculture*, vol. 596, Feb. 2025, Art. no. 741719, doi: 10.1016/j.aquaculture.2024.741719.

[6] E. J. Livengood, C. L. Ohs, and F. A. Chapman, *Candidate Species for Florida Aquaculture: Discus Symphysodon spp., a Profitable but Challenging Species for Florida Aquaculture*. Gainesville, FL, USA: University of Florida IFAS Extension, rev. 2024. Available at: https://journals.flvc.org/edis/article/view/118440

[7] S.-C. Zhao, B. Wen, J.-Z. Gao, and Z.-Z. Chen, "Proteomic analysis revealed gender-related differences in the skin mucus proteome of discus fish (Symphysodon haraldi) during the parental and non-parental care periods," Comp. Biochem. Physiol. Part D: Genomics Proteomics, vol. 52, Dec. 2024, Art. no. 101314, doi: 10.1016/j.cbd.2024.101314.

[8] D. da C. Mattos, L. D. Cardoso, A. T. de Oliveira, R. Screnci-Ribeiro, B. O. de Mattos, P. H. R. Aride, M. C. Radael, J. H. de S. Motta, and M. V. Vidal, "Effect of temperature on the embryonic and larvae development of discus fish Symphysodon aequifasciatus and time of first feeding," Zygote, vol. 32, no. 4, pp. 279–284, Aug. 2024, doi: 10.1017/S0967199424000236.

[9] D. Li and L. Du, "Recent advances of deep learning algorithms for aquacultural machine vision systems with emphasis on fish," Artif. Intell. Rev., vol. 55, no. 5, pp. 4077–4116, Jun. 2022, doi: 10.1007/s10462-021-10102-3.

[10] D. An, J. Hao, Y. Wei, Y. Wang, and X. Yu, “Application of computer vision in fish intelligent feeding system—A review,” Aquac. Res., vol. 52, no. 2, pp. 423–437, Feb. 2021, doi: 10.1111/are.14907.

[11] E. D’Agaro, “Fish Farming 5.0: Advanced Tools for a Smart Aquaculture Management,” Appl. Sci., vol. 15, no. 23, p. 12638, Nov. 2025, doi: 10.3390/app152312638.

[12] S. Al-Abri, S. Keshvari, K. Al-Rashdi, R. Al-Hmouz, and H. Bourdoucen, “Computer vision based approaches for fish monitoring systems: a comprehensive study,” Artif. Intell. Rev., vol. 58, no. 6, Mar. 2025, Art. no. 185, doi: 10.1007/s10462-025-11180-3.

[13] Y.-P. Huang and S. P. Khabusi, “Artificial Intelligence of Things (AIoT) advances in aquaculture: A review,” *Processes*, vol. 13, no. 1, p. 73, Jan. 2025, doi: 10.3390/pr13010073.

[14] H. Tamut, R. Ghosh, K. Gosh, and M. A. S. Siddique, “Enhancing Disease Detection in the Aquaculture Sector Using Convolutional Neural Networks Analysis,” Aquac. J., vol. 5, no. 1, Mar. 2025, Art. no. 6, doi: 10.3390/aquacj5010006.

[15] A. Fitzgerald, C. C. Ioannou, S. Consuegra, A. Dowsey, and C. G. de Leaniz, “Machine vision applications for welfare monitoring in aquaculture: Challenges and opportunities,” *Aquac. Fish Fish.*, vol. 5, no. 1, Jan. 2025, Art. no. e70036, doi: 10.1002/aff2.70036.

[16] Y. Fu, Z. Xu, B. Wen, J. Gao, and Z. Chen, “Gonad-Specific Transcriptomes Reveal Differential Expression of Gene and miRNA Between Male and Female of the Discus Fish (Symphysodon aequifasciatus),” Front. Physiol., vol. 11, Aug. 2020, Art. no. 754, doi: 10.3389/fphys.2020.00754.

[17] Seneye, “Aquarium monitor system – fish tank water sensor.” [Online]. Available: https://www.seneye.com/. [Accessed: Mar. 18, 2026].

[18] Neptune Systems, “Apex Fusion.” [Online]. Available: https://www.neptunesystems.com/apex-fusion/. [Accessed: Mar. 18, 2026].

APPENDICES

The appendices of this individual proposal contain selected supporting materials that provide additional technical, design, and planning detail for the proposed Discus breeding lifecycle support module.

Table 9.1:Summary of Appendices

Appendix	Content	Purpose
Appendix A	Survey instruments	Includes sample stage-specific user input questions for pre-spawning, egg-stage, and fry-stage support
Appendix B	Detailed diagrams	Includes the methodology workflow diagram and the high-level system architecture diagram
Appendix C	Preliminary UI mockups	Includes proposed interface screens for stage selection, image upload, context input, and result display
Appendix D	Risk analysis	Includes key technical and implementation risks together with mitigation strategies
Appendix E	Ethical documents	Includes ethical handling notes related to non-invasive data collection, responsible image use, and decision-support limitations

Appendix A: Survey Instruments
This appendix contains sample structured questions used to support stage-specific input collection.
Example questions

- Have two Discus fish been repeatedly staying together?
- Have the fish shown repeated surface-cleaning behavior?
- Are the parents fanning the eggs?
- Are any eggs visibly white or opaque?
- Are fry visibly clustered on or near the parents?
- Have fry started responding to feed?

Appendix B: Detailed Diagrams
This appendix contains the main supporting diagrams used in the proposal.
Included figures

- Figure 3.3: Overall User Workflow of the Proposed Discus Breeding Lifecycle Support Module
- Figure 4.1: High-Level System Architecture of the Proposed Individual Module

Appendix C: Preliminary UI Mockups
This appendix contains early interface concepts for the proposed module.
Included screens

- stage selection
- image input

- contextual input
- result and warning display
- breeding-cycle record view

Appendix D: Risk Analysis
This appendix contains the main implementation risks identified for the proposed module.
Key risks

- poor image quality
- limited dataset availability
- incorrect user input
- missing lifecycle records
- model overfitting

Appendix E: Ethical Documents
This appendix contains short ethical handling notes for the proposed system.
Included notes

- non-invasive data collection only
- responsible handling of breeder-provided images and records
- data privacy and secure record handling
- advisory nature of system outputs